

Practical Quantum Subspace Expansion: Stability, Noise, and Scaling

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Abstract

Keywords

quantum subspace expansion; variational quantum eigensolver; excited states; active spaces; measurement noise

Abbreviations

QSE, quantum subspace expansion; VQE, variational quantum eigensolver; UCC, unitary coupled cluster; CASCI, complete active space configuration interaction.

Introduction

The Quantum Computing Edge in Quantum Chemistry

1. Discuss why quantum computing has an edge in quantum chemistry. Relate to the exponential scaling that classical computers will face. 2. Talk about how quantum computers will also not be perfect, mainly limited by statistical noise and hardware errors (lack of error correcting code). + scaling also can be rough even if its not as bad as classical 3. Discuss the various advances on generating excited states (briefly explain why we are interested in it). Mainly need to answer the question, WHY subspace expansion? Why QSE?

Quantum Subspace Expansion for Excited States

Add a diagram for workflow?

Quantum Subspace Expansion is a subspace exploration method, that builds a subspace around a guess ansatz. This is typically done by first obtaining a ground state reference state via a Variational Quantum Eigensolver (VQE). This is another hybrid-quantum method that uses a classical optimiser to optimise coefficients of circuit parameters.

In this study, the Variational Quantum Eigensolver (VQE) algorithm was employed to approximate a ground reference state $\psi_V QE$. 6 different classes of Unitary Coupled Cluster (UCC) Ansatz were applied, namely UCCSD, UCCSDSinglet, UCCGSD and kUpCCGSDSinglet ($k = 1, 2, 3$). Each Ansatz is characterised by a unitary Operator defined as

$$\hat{U}_{\text{UCC}}(\boldsymbol{\theta}) = \exp\left(\hat{T}(\boldsymbol{\theta}) - \hat{T}^\dagger(\boldsymbol{\theta})\right) = \exp\left[\sum_{\lambda} \theta_{\lambda} \left(\hat{T}_{\lambda} - \hat{T}_{\lambda}^\dagger\right)\right]. \quad (1)$$

whereby $\hat{T}$ is a parameterised excitation operator. Under the exponent, its self adjoint $\hat{T}^\dagger$ is subtracted, rendering the UCC operator unitary. The UCC operator was applied to the single reference Hartree Fock (HF) state to get the ground reference state.

$$|\Psi_{\text{UCC}}(\boldsymbol{\theta})\rangle = \hat{U}_{\text{UCC}}(\boldsymbol{\theta}) |\Phi_{\text{HF}}\rangle. \quad (2)$$

Given an optimized reference state $|\Psi_0\rangle$, QSE constructs a non-orthogonal projected basis by applying a set of expansion operators $\{\hat{O}_i\}$,

$$|\Phi_i\rangle = \hat{O}_i |\Psi_0\rangle. \quad (3)$$

The Hamiltonian and overlap matrices in this projected basis are

$$H_{ij} = \langle \Phi_i | \hat{H} | \Phi_j \rangle = \langle \Psi_0 | \hat{O}_i^\dagger \hat{H} \hat{O}_j | \Psi_0 \rangle, \quad (4)$$

$$S_{ij} = \langle \Phi_i | \Phi_j \rangle = \langle \Psi_0 | \hat{O}_i^\dagger \hat{O}_j | \Psi_0 \rangle. \quad (5)$$

The QSE energies are obtained by solving the generalized eigenvalue problem

$$\mathbf{H}\mathbf{c} = E\mathbf{S}\mathbf{c}. \quad (6)$$

In practice, the overlap matrix is first diagonalized as

$$\mathbf{S} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\dagger, \quad \mathbf{V}^\dagger\mathbf{V} = \mathbf{I}. \quad (7)$$

This defines the orthogonalizing transformation

$$\mathbf{X} = \mathbf{V}\mathbf{\Lambda}^{-1/2}, \quad \mathbf{X}^\dagger\mathbf{S}\mathbf{X} = \mathbf{I}. \quad (8)$$

Using $\mathbf{c} = \mathbf{X}\mathbf{y}$, the generalized eigenvalue problem is converted into an ordinary Hermitian eigenvalue problem,

$$\mathbf{X}^\dagger\mathbf{H}\mathbf{X}\mathbf{y} = E\mathbf{y}, \quad \mathbf{c} = \mathbf{X}\mathbf{y}. \quad (9)$$

The eigenvector matrix is first multiplied by the whitening factor $1/\sqrt{\lambda_i}$. Crucially, this means that when λ_i is small, this factor becomes large. As such, noise and errors in the sampled matrix elements can be amplified greatly. This is the heart of what many consider the instability problem that plagues the generalized eigenvalue problem. This problem cannot be easily ignored because statistical sampling noise is inherent to quantum computers.

Therefore, in practice, thresholding is commonly used. Tiny overlap eigenvalues below a threshold τ are often dropped to prevent numerical instabilities. If $\mathbf{V}_r$ and $\mathbf{\Lambda}_r$ contain only the retained eigenvectors and eigenvalues with $\lambda_i > \tau$, the solved problem becomes

$$\mathbf{X}_r^\dagger\mathbf{H}\mathbf{X}_r\mathbf{y} = E\mathbf{y}, \quad \mathbf{X}_r = \mathbf{V}_r\mathbf{\Lambda}_r^{-1/2}. \quad (10)$$

This reduces the number of solutions that can be recovered and lowers the effective rank of the QSE subspace.

Current QSE Studies and Contributions of This Work

Results and Discussion

Effective Dimensions of the QSE Subspace

Subspace Completeness Beyond Ansatz Choice

Finite-Shot Effects on Generalized Eigenvalue Stability

Scaling of QSE and Future Adaptive Strategies

Computational Methods

Molecular Geometries and Active Space Preparation

The dataset used comprised 28 closed-shell organic molecules from Thiel’s excited-state benchmark set [1]. The published Cartesian nuclear coordinates were referenced directly and used without further optimisation.

For each molecule, Restricted Hartree-Fock calculations were performed using PySCF [2, 3] with the STO-3G basis set. This set of canonical molecular orbitals was optimised once and used consistently throughout all subsequent CASCI, VQE and QSE calculations in the study.

Active spaces (denoted $n_e n_o$) comprised the $n_e/2$ occupied molecular spatial-orbitals closest to the HOMO and the $n_o - n_e/2$ virtual molecular spatial-orbitals closest to the LUMO. The remaining occupied orbitals were treated as frozen and doubly occupied, while the remaining virtual orbitals were excluded from calculations.

Exact Reference

Within each active space, the exact reference energies and wavefunctions were obtained from complete active-space configuration interaction (CASCI) calculations performed using PySCF. Calculations were separated by spin-projection sector M_S by setting the number of active α and β spin electrons to be

$$n_\alpha = \frac{n_e}{2} + M_S, n_\beta = \frac{n_e}{2} - M_S$$

Singlet states were obtained from the $M_S = 0$ sector, while triplet states were calculated in the $M_S = 0, +1, -1$ sectors. For each active-space spin projection sector, the determinant-space dimension, and therefore the maximum number of available roots, was $\binom{n_o}{n_\alpha} \binom{n_o}{n_\beta}$.

Where available, the lowest 11 singlet states (ground state + 10 excited states), and the lowest 10 triplet states (within each spin projection sector) were calculated. To ensure that the target number of states was obtained, three times the required number of CASCI roots were initially requested from the FCI solver, subject to the maximum dimension of the corresponding determinant space. The resulting roots were then filtered according to their $\langle \hat{S}^2 \rangle$ expectation values, retaining only roots that satisfy $|\langle \hat{S}^2 \rangle - S(S+1)| < 10^{-5}$ with $S = 0$ for singlets and $S = 1$ for triplets.

Throughout this study, wavefunctions were represented in a common active spin-orbital determinant basis by a state vector of length $\binom{2n_o}{n_e}$. This vector includes all determinants with n_e electrons distributed among the $2n_o$ active spin-orbitals (irrespective of their spin projections). Determinants were ordered lexicographically using an interleaved spin-orbital convention ($1\alpha, 1\beta, 2\alpha, 2\beta, \dots$).

PySCF represented each wavefunction as a two-dimensional CI coefficient matrix with dimensions $\binom{n_o}{n_\alpha}$ by $\binom{n_o}{n_\beta}$, corresponding to the α and β electron occupation strings. These coefficients were embedded into the corresponding entries of the common state-vector representation. During this conversion, the fermionic phase arising from the permutation between PySCF's separated α and β electron ordering and the interleaved spin-orbital convention was included explicitly.

Quantum Computing Framework

All quantum-circuit calculations in this study were performed using classical simulations implemented with the Qibo and Qibochem [4, 5] frameworks. These frameworks were used to construct, represent and simulate the quantum circuits employed. A research fork of Qibochem was developed to support the computational workflow used in the study. The fork introduced configurable ansatz classes, optimised circuit evaluation routines, support for measuring spin observables, and dedicated functionality for constructing and sampling the projected matrices required for Quantum Subspace Expansion calculations.

All calculations in this work used ideal noiseless circuit simulations without modelling any hardware. Only finite-sampling noise arising from a limited number of measurement shots was considered in the relevant sections.

Hamiltonian and Spin Operators

For each active space, the electronic Hamiltonian was constructed in second quantisation as follows:

$$\hat{H} = E_{\text{nuc}} + \sum_{pq} h_{pq} \hat{a}_p^\dagger \hat{a}_q + \frac{1}{2} \sum_{pqrs} g_{pqrs} \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r. \quad (11)$$

where E_{nuc} is the core nuclear repulsion energy and p, q, r, s index the active spin orbitals. h_{pq} and g_{pqrs} are the one and two-electron integral coefficients evaluated in the fixed canonical Hartree-Fock MO basis.

Spin operators were constructed in the same second quantisation basis as follows.

$$\hat{S}_z = \frac{1}{2} \sum_p (\hat{n}_{p\alpha} - \hat{n}_{p\beta}), \quad \hat{n}_{p\sigma} = \hat{a}_{p\sigma}^\dagger \hat{a}_{p\sigma}. \quad (12)$$

$$\hat{S}_+ = \sum_p \hat{a}_{p\alpha}^\dagger \hat{a}_{p\beta}, \quad \hat{S}_- = \sum_p \hat{a}_{p\beta}^\dagger \hat{a}_{p\alpha}. \quad (13)$$

$$\hat{S}^2 = \hat{S}_z^2 + \frac{1}{2} (\hat{S}_+ \hat{S}_- + \hat{S}_- \hat{S}_+). \quad (14)$$

All fermionic Hamiltonian, spin and excitation operators were mapped to qubit operators using the Jordan-Wigner (JW) transformation.

Ground State Reference

As introduced above, approximate ground-state reference wavefunctions were prepared using parameterised Unitary Coupled-Cluster (UCC) Ansatz. In this study, six ansatz families were used, namely: UCCSD, UCCSDSinglet, UCCGSD, and k -UpCCGSDSinglet with $k = 1, 2$, and 3. The Ansatz were characterised by the excitation generators included in the cluster operator $\hat{T}$, summarised in the following table.

Table 1: Classification of the 6 UCC ansatz families

Ansatz	Singles excitations	Doubles excitations	Spin- conserving	Generalised	Spin- adapted	Paired
UCCSD	✓	✓	✓			
UCCSDSinglet	✓	✓	✓		✓	
UCCGSD	✓	✓	✓	✓		
1UpCCGSDSinglet	✓	✓	✓	✓	✓	✓
2UpCCGSDSinglet	✓	✓	✓	✓	✓	✓
3UpCCGSDSinglet	✓	✓	✓	✓	✓	✓

For each ansatz, candidate singles and doubles transitions were first enumerated in fermionic form. Only canonical transitions that excite from a lower to higher spin orbital were retained because the reverse transitions would already be included through $\hat{T}^\dagger$ before exponentiation. All retained generators were spin conserving: single excitations preserved spin labels, while double excitations preserved the total number of α and β spin electrons.

The non-generalised UCCSD and UCCSDSinglet ansatz were restricted to occupied-to-virtual single and double excitations relative to the reference Hartree-Fock determinant [6]. In contrast, the generalised UCCGSD ansatz allowed generalised orbital transitions between occupied-occupied and virtual-virtual spin orbitals [7, 8]. To reduce the size of the generalised-double operator pool, mixed-direction double excitations, in which one electron was promoted while the other was relaxed, were excluded. Across the *4e4o* to *8e8o* active spaces, this practical restriction reduced the number of independent parameters by 42.2–45.2% and the circuit depth by 46.4–47.6%.

The k -UpCCGSDSinglet ansatz combined generalised singles with generalised paired doubles. For each paired double excitation, an $\alpha\beta$ electron pair was removed from one spatial orbital and created in another. The UCCSDSinglet and k -UpCCGSDSinglet constructions imposed spin-adapted parameter sharing between spin-orbital excitations that had the same spatial-orbital labels. For k -UpCCGSDSinglet, the complete generalised single and paired-double block was repeated k times, with an independent set of variational parameters assigned to each repetition.

Unless otherwise stated, all UCC operators were implemented using the first-order Suzuki Trotter product approximation given in eq. (15).

$$e^{\hat{T} - \hat{T}^\dagger} \approx \prod_{\lambda} e^{\theta_{\lambda} (\hat{T}_{\lambda} - \hat{T}_{\lambda}^\dagger)}. \quad (15)$$

At every optimisation step, the circuit energy was evaluated as the noiseless statevector expectation value of the electronic Hamiltonian eq. (11). The variational parameters were optimised classically using the L-BFGS-B optimiser through Qibo, to minimise the ground state expectation value. For the k -UpCCGSDSinglet ansatz, the parameters associated with all k repeated layers were optimised simultaneously. Before optimisation, ansatz parameters were initialised based on the following hierarchy:

$$\text{MP2} \rightarrow \text{1-UpCCGSD} \rightarrow \text{2-UpCCGSD} \rightarrow \text{3-UpCCGSD} \rightarrow \text{UCCSDSinglet} \rightarrow \text{UCCSD} \rightarrow \text{UCCGSD}.$$

Parameters associated with excitation generators shared with the preceding ansatz were initialised with the previously optimised amplitudes, while newly introduced generators were initialised to zero. The initial occupied-to-virtual double-excitation parameters were estimated using second-order Moller-Plesset perturbation theory (MP2). For parameter transfer from a k -UpCCGSDSinglet ansatz to a subsequent ansatz, the optimized amplitudes associated with each generator were summed across the k layers and used as the initial amplitude for the corresponding generator

Following optimisation, the resulting VQE parameters were stored as a fixed ground-state reference wavefunction. This fixed reference set was used consistently for all subsequent statevector and finite-shot QSE calculations.

QSE Operator Pools and Projected Matrices

As discussed in the introduction, separate one-body operator pools were used to construct the singlet and triplet-targeting QSE subspaces. The singlet operator pool, $\mathcal{O}^S$, was defined as

$$\hat{O}_{ij}^S = \hat{a}_{i\alpha}^\dagger \hat{a}_{j\alpha} + \hat{a}_{i\beta}^\dagger \hat{a}_{j\beta}, \quad (16)$$

$$\mathcal{O}^S = \left\{ \hat{O}_{ij}^S \mid i, j \in \{1, \dots, n_o\} \right\}. \quad (17)$$

whereby i and j to denote active spatial orbitals.

The triplet operator pool, $\mathcal{O}^T$, was constructed from all three spin-projection components, which were collected into a single operator pool.

$$\hat{O}_{ij}^{T,0} = \frac{1}{\sqrt{2}} \left(\hat{a}_{i\alpha}^\dagger \hat{a}_{j\alpha} - \hat{a}_{i\beta}^\dagger \hat{a}_{j\beta} \right), \quad (18)$$

$$\hat{O}_{ij}^{T,+1} = \hat{a}_{i\alpha}^\dagger \hat{a}_{j\beta}, \quad \hat{O}_{ij}^{T,-1} = \hat{a}_{i\beta}^\dagger \hat{a}_{j\alpha}, \quad (19)$$

$$\mathcal{O}^T = \left\{ \hat{O}_{ij}^{T,m} \mid i, j \in \{1, \dots, n_o\}, m \in \{-1, 0, +1\} \right\}. \quad (20)$$

The singlet operator pool therefore contains n_o^2 operators, while the triplet pool contains $3n_o^2$ operators. Individual pool elements were enumerated as $\{\hat{O}_\mu\}$, with μ and ν indexing arbitrary operators within the given pool.

A projected observable matrix were constructed for the overlap, Hamiltonian and spin-squared matrices. Observable matrix elements were defined as follows:

$$\bar{s}_{\mu\nu} = \hat{O}_\mu^\dagger \hat{O}_\nu, \quad \bar{h}_{\mu\nu} = \hat{O}_\mu^\dagger \hat{H} \hat{O}_\nu, \quad \bar{z}_{\mu\nu} = \hat{O}_\mu^\dagger \hat{S}^2 \hat{O}_\nu. \quad (21)$$

where the corresponding target observables were $\hat{I}$, $\hat{H}$, and $\hat{S}^2$, respectively.

The fermionic QSE expansion operators and each target observable were mapped separately to Pauli operators using the Jordan-Wigner transformation. The projected observables were then constructed by multiplication in the Pauli representation. Terms with coefficients of magnitude smaller than 10^{-12} were discarded after the final product construction. The resulting Pauli-string expectation values were then evaluated with respect to the optimised VQE states, $|\Psi_{\text{VQE}}\rangle$. For all calculations performed, only the diagonal and upper-triangular matrix elements, $\mu \leq \nu$ were constructed and evaluated. Lower-triangular matrix terms were then reconstructed using Hermitian symmetry.

Finite-Shot Sampling

Finite-shot sampling was used to model the effects of statistical measurement noise on the projected Hamiltonian and overlap matrices. For each calculation, all unique Pauli strings in the projected Hamiltonian and overlap observable matrices were first collected into a global measurement pool. Then, a largest-first greedy qubit-wise-commuting (QWC) grouping algorithm was used to partition the global measurement pool. Grouping was performed globally across elements from both matrices which reduced the total number of commuting groups that needed to be measured. Preliminary data also showed that this method of global grouping, as opposed to grouping terms for each matrix element, resulted in much better agreement with the statevector results.

For each commuting group, a separate measurement circuit was constructed by appending basis-rotation gates to the optimized VQE circuit, such that all Pauli strings in the group could be measured in the computational (Z) basis. Hadamard gates were applied to qubits measured in the (X) basis, while $S^\dagger$ followed by a Hadamard gate was applied to qubits measured in the (Y) basis. This produced a distinct rotated statevector for each commuting group.

Measurement outcomes were generated for each rotated statevector using NumPy's multinomial sampler. The expectation values of all Pauli strings within a commuting group were estimated from the resulting sampled frequencies. These expectation values were then used to reconstruct the Hamiltonian and overlap matrices.

At each specified shot count, the finite-shot calculation was repeated using ten statistically independent sampling iterations. The resulting Hamiltonian and overlap matrices were solved independently, and the reported energy errors were calculated from the mean of the ten independently obtained energies.

Numerical Analysis and Evaluation Metrics

Generalised Eigenvalue Problem

Due to the non-orthogonality of the QSE basis states, solving for the energy eigenvalues and wavefunctions involves solving a generalised eigenvalue problem $Hc = ES c$. This was done by first diagonalising the Hermitian overlap matrix S .

$$\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\dagger. \quad (22)$$

where the columns of $\mathbf{U}$ are the eigenvectors of $\mathbf{S}$ and $\mathbf{\Lambda}$ is a diagonal matrix containing the eigenvalues of $\mathbf{S}$. Then, we define the whitening factor

$$\mathbf{V} = \mathbf{U}\mathbf{\Lambda}^{-1/2}, \quad \mathbf{V}^\dagger \mathbf{S} \mathbf{V} = \mathbf{I}. \quad (23)$$

Using $c = \mathbf{V}\mathbf{y}$, we get

$$\mathbf{V}^\dagger \mathbf{H} \mathbf{V} \mathbf{y} = E \mathbf{V}^\dagger \mathbf{S} \mathbf{V} \mathbf{y} = E \mathbf{y}. \quad (24)$$

Numerical instabilities arise when the eigenvalues used are tiny or near-null, because noise perturbations will be greatly amplified by a factor $\frac{1}{\sqrt{\lambda}}$. Therefore in practice, the following reduced eigenvalue equation is solved

$$\tilde{\mathbf{V}}^\dagger \mathbf{H} \tilde{\mathbf{V}} \mathbf{y} = E \mathbf{y}. \quad (25)$$

where $\tilde{\mathbf{V}}$ and $\tilde{\mathbf{V}}^\dagger$ denote a reduced form of $\mathbf{V}$ and $\mathbf{V}^\dagger$, retaining only a selected subset of its columns. This gives the wavefunction in the QSE operator basis as $\mathbf{c}_k = \tilde{\mathbf{V}}\mathbf{y}_k$.

For all statevector calculations used in this work, the retained subset was chosen from a fixed maximum dimension determined by the active space and spin sector as follows.

$$\begin{aligned} d_S^{\max} &= \min \left[\binom{n_o}{n_p}^2 - \binom{n_o}{n_p+1} \binom{n_o}{n_p-1}, n_o^2 \right], \\ d_T^{\max} &= \min \left[3 \left(\binom{n_o}{n_p+1} \binom{n_o}{n_p-1} - \binom{n_o}{n_p+2} \binom{n_o}{n_p-2} \right), 3n_o(n_o-1) \right], \quad n_p = \frac{n_e}{2}. \end{aligned} \quad (26)$$

The first term in each expression gives the maximum number of physically distinct singlet or triplet roots available within the corresponding active space. These values were obtained from differences between the dimensions of adjacent fixed- M_S determinant sectors, such that

$$N_S = D_{M_S=S} - D_{M_S=S+1}$$

This subtraction isolates the number of spin- (S) multiplets because the $(M_S = S)$ sector contains one component from every state with total spin (S) or greater, whereas the $(M_S = S + 1)$ sector contains only components with total spin greater than (S) .

For the $2e2o$, $2e3o$, and $4e3o$ active spaces, the number of physically available roots was smaller than the nominal number of QSE expansion operators. The retained dimension was therefore capped by the corresponding physical root count. This cap was in place because it was noted empirically that additional operator directions beyond the physically available space produced near-null eigenvalues. More generally, this restriction applies to active spaces that do not include orbitals extending from HOMO-1 through LUMO+1.

For the triplet subspace, the retained dimension was further limited to $(3n_o(n_o - 1))$. The triplet number operators were extremely weakly supported by the paired reference states and generated overlap eigenvalues that could not be resolved reliably above numerical precision. For consistency across ansatz, these directions were therefore excluded for all triplet statevector calculations.

For finite-shot calculations, a threshold τ was used to define the retained subspace. Only eigenvectors with eigenvalues satisfying $\lambda_i > \tau$ were retained. The thresholds τ were drawn from a fixed logarithmic grid from 10^{-4} to 10^0 in increments of 0.1 in the exponent.

Spin-expectation Evaluation

For the k^{th} recovered state, the spin expectation value was calculated as follows:

$$\langle \hat{S}^2 \rangle_k = \frac{\mathbf{c}_k^\dagger \mathbf{Z} \mathbf{c}_k}{\mathbf{c}_k^\dagger \mathbf{S} \mathbf{c}_k}. \quad (27)$$

Using $\mathbf{c}_k = \tilde{\mathbf{V}} \mathbf{y}_k$ and $\tilde{\mathbf{V}}^\dagger \mathbf{S} \tilde{\mathbf{V}} = \mathbf{I}$, this expression becomes

$$\langle \hat{S}^2 \rangle_k = \frac{\mathbf{y}_k^\dagger \tilde{\mathbf{V}}^\dagger \mathbf{Z} \tilde{\mathbf{V}} \mathbf{y}_k}{\mathbf{y}_k^\dagger \mathbf{y}_k}. \quad (28)$$

Projection

TBC results first

Triplet Degeneracies

TBC results first

Non Index State Matching

TBC results first

Conclusions

Acknowledgements

Data Availability

Supporting Information

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